

End-Term Semester Examination 2025

Paper set: 2

Name of the program: **B. Tech**

Name of the course: **Animal Biotechnology**

Semester: **VI**

Course code: **TBT-601**

Time: 3.0 hrs

Maximum Marks: 100

- Q. 1: a) Highlight basic technology of animal cell culture and their applications (CO-1)
b) Define importance of serum quality in cell culture applications
c) Sketch-out methods for cryopreservation and transport of animal germplasm
- Q. 2: a) Explain role of DNA microinjection in transgenic technology (CO-2)
b) Explain different retroviral vectors on trans-genetics and cell therapies
c) Implement technique of *in vitro* fertilization and embryo transfer in Infertility/transgenic breeding
- Q. 3: a) Explain basic concept of gene therapy and its current usage (CO-3)
b) Sketch-out basic principle of transfection and methodology in use.
c) Highlight different biological techniques for rapid diagnosis of genetic disorders
- Q. 4: a) Highlight basic principle of gene-clonning and expression vectors (CO-4)
b) Analyze extinction of gene function based on several techniques currently in use
c) Highlights major differences among prokaryotic and eukaryotic gene regulation mechanisms
- Q. 5: a) Sketch-out general principles of nutrition and importance of sea-foods (CO-5)
b) Highlight basic concept of marine biotechnology and their importance of human health
c) Analyze methodology behind the production of transgenic fishes and their importance to marine ecology.

